Composing the error budget of an optical clock: one rate scalar per atom

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An optical clock's systematic error budget is built today as a table, one row per physical effect: blackbody radiation, lattice or probe Stark shifts, the electric-quadrupole shift, motional time dilation, gravitational redshift. Each row is computed in its own formalism. The rows are summed linearly, and their uncertainties are combined in quadrature, the square root of the sum of their squares. Fringe contrast is measured separately, as an experiment outside the budget's own rows. We describe an alternative already implemented and validated in the open-source CliffordClock pipeline. Every systematic enters as a multiplicative factor on a single scalar rate, the rate at which an atom's own proper time advances along its worldline. The accumulated phase is the time integral of that rate. The clock's reported fractional shift is the mean of the resulting ensemble phase distribution. The Ramsey fringe visibility is the modulus of the mean phasor, fixed by that same distribution's phase variance under the standard Gaussian-phase closure, the assumption that the ensemble's accumulated phases are Gaussian-distributed. The additive table is the first-order term of this product's own expansion. We give the pipeline's own bounds on the terms it drops: a DC-Stark×blackbody product correction of order 10^{-33}, and a gravitational×field-driven cross term of order 10^{-31}. Both bounds sit at least twelve orders of magnitude below the 10^{-18}-10^{-19} systematic floor of the clocks the budget describes. The value of the product form is structural. It changes no published total. We show three consequences of the composed form, using numbers already validated in the companion paper. First, a multi-surface thermal-environment term needs its higher-order moments evaluated per surface, before they are combined. Second, a trapped-ion secular-motion reconciliation composes two systematics as correlated factors on one rate, closing a discrepancy of 14σ down to 0.08σ against a published budget. Third, a Ramsey visibility comes from the same worldline ensemble that sets the mean shift, with no separate fit required. Relativistic timing already composes rate scalars for its own gravitational and kinematic terms. This paper's claim is one of scope. It extends the same composition to the atomic-environment terms of a clock's full budget, evaluated per quantum state. Fringe contrast comes out of the phase variance of that same object. The implementation is open and checked against published budgets. Code, tests, and the six-line composition rule are public under the AGPLv3.

I. INTRODUCTION

A modern optical clock's systematic-uncertainty budget is built one row at a time, and each row draws on its own kind of data. The blackbody-radiation shift comes from a Planck-integral model of the clock's radiative environment. The lattice or probe AC-Stark shift comes from the trap-light polarizability. The DC-Stark shift and its spatial gradient come from a field survey. The electric-quadrupole shift, present for an ion with $J \geq 1$, comes from the local field-gradient tensor and an atomic theory coefficient. The motional time-dilation shift comes from sideband-thermometry-measured mode populations. The gravitational redshift comes from a height survey against a local reference [1]. Each row is a physically independent calculation, in its own formalism, against its own inputs. The rows are then summed linearly to a total fractional shift, and their uncertainties are combined in quadrature to a total systematic uncertainty. The clock's fringe contrast, the visibility that sets how well the interrogation resolves the transition, is measured separately, as an experimental observation outside the budget's own rows.

Two recent publications make this practice concrete with real numbers. Aeppli et al. report an ^{87}Sr optical lattice clock's blackbody shift as $-4.841720 \times 10^{-15}$, with a stated uncertainty of $\pm 7.3 \times 10^{-19}$, from a Planck-integral model of the clock's measured in-vacuum temperature combined with fitted dynamic polarizability coefficients [2]. Their total systematic uncertainty, after every such row is combined, is 8×10^{-19} [2]. Bothwell et al., on the same platform, publish a DC-Stark gradient row of $+0.3(0.2) \times 10^{-20}$ per millimetre from a field survey. In the same table, a gravitational-redshift slope from two independent analysis methods gives $-9.8(2.3) \times 10^{-20}$ and $-12.8(2.7) \times 10^{-20}$ per millimetre respectively. The two methods differ from each other by 0.85σ, even though both describe the same physical effect [3]. On a different platform and a different species, Brewer et al. report an ^{27}Al$^+$/^{25}Mg$^+$ trapped-ion clock's own motional time-dilation row as $-17.3(2.9) \times 10^{-19}$, one line among the rows that sum to their clock's total

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systematic uncertainty, published as below 10^{-18} [4]. A later evaluation of the same species pair on a different trap, Marshall et al., reports the same row at $-114.6(3.8) \times 10^{-19}$ and a total systematic uncertainty of 5.5×10^{-19} [5]. Four papers, two atomic species, two very different orders of magnitude for nominally the same physical row: the practice that produces these numbers is precise and well established. It is a practice, and this paper asks what one composed object, replacing the table of separately computed rows, would look like, and where it would matter.

CliffordClock is an open-source pipeline, described and validated against published clock budgets in a companion paper,[1] that already implements the alternative this paper describes. Every systematic in that pipeline, DC-Stark, blackbody radiation, the electric-quadrupole shift, motional time dilation, gravitational redshift, enters as a multiplicative factor on one scalar rate at which an atom's own proper time advances. The clock's reported shift and its fringe visibility both come out of the same accumulated-phase object. Section II states that composition rule. Section III derives the additive table as its first-order expansion and gives the pipeline's own bound on what that expansion drops. Section IV shows three places, already validated against published numbers, where the fuller composition changes what a lab would conclude. Section V places this against relativistic-timing practice, which already composes rate factors for gravitational and kinematic terms. Section VI gives the composition rule in six lines of code.

II. THE COMPOSITION LAW

This section states the composition rule, and shows how the clock's reported shift and its fringe visibility both come from the same object. Write $\nu(t)$ for the clock transition's instantaneous frequency along one atom's worldline, its own path through space, time, and internal quantum state. In the standard treatment, each systematic k contributes an independently computed fractional perturbation p_k, and the rows are added: $\nu(t)/\nu_0 - 1 \approx \sum_k p_k$. The composition rule this paper describes instead multiplies:

$$\nu(t) \;=\; \nu_0 \prod_k \big(1 + p_k(x(t))\big), \tag{1}$$

where $x(t)$ is the atom's full state along its worldline at time t, position, velocity, and any internal or motional quantum number each term's own physics depends on. Equation (1) is the local proper-time rate: general relativity already has names for a scalar factor multiplying an ideal clock's rate this way, the lapse function in a 3+1 decomposition of spacetime, or, for a static observer, the redshift factor $\sqrt{-g_{00}}$. Each term p_k is still computed by its own physics. The DC-Stark shift comes from the transition's differential polarizability and the local field. The blackbody shift comes from a Planck-integral model of the radiative environment. The quadrupole shift comes from the local field-gradient tensor. The motional shift comes from the atom's own velocity or vibrational occupation. The gravitational term comes from height in a local potential. Nothing about any individual term changes. What changes is how the terms are put back together. Equation (1) multiplies them as factors on one shared rate. A budget table instead adds them, one row per systematic, an addition Sec. III shows is this product's own first-order truncation.

An atom's accumulated phase over an interrogation of proper duration T is the time integral of that rate,

$$\Delta\Phi \;=\; 2\pi \int_0^T \nu(t)\, dt, \tag{2}$$

in radians. An ensemble of M atoms, each with its own worldline (its own trajectory, motional state, and local field environment), gives M accumulated phases $\Delta\Phi_1, \ldots, \Delta\Phi_M$. The clock's reported fractional frequency shift is the ensemble mean of these phases, normalized by the unperturbed phase $2\pi\nu_0 T$:

$$\frac{\Delta\nu}{\nu_0} \;=\; \frac{\langle \Delta\Phi \rangle}{2\pi\nu_0 T} - 1, \tag{3}$$

the first moment of the ensemble's phase distribution, converted to a fractional frequency. The Ramsey fringe visibility, the same contrast a lab measures at the end of an interrogation, is the modulus of the mean phasor over that same ensemble,

$$V \;=\; \big|\langle e^{i\Delta\Phi} \rangle\big| \;\le\; 1. \tag{4}$$

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[1] The companion paper is `paper/main.tex` in the project repository, `https://github.com/velar-mbr/CliffordClock`. Every case-study number in this paper is copied from that paper, from `docs/CONVENTIONS.md`, or from `notebooks/13_trapped_ion_quantum_motion.ipynb`; this paper's own text does not recompute any of them.

For phases that are Gaussian-distributed across the ensemble, the characteristic-function identity $\langle e^{i\Delta\Phi}\rangle = e^{i\langle\Delta\Phi\rangle - \sigma_\Phi^2/2}$ gives the closed form $V = \exp(-\sigma_\Phi^2/2)$ exactly. The visibility is fixed by the phase distribution's variance, the same distribution whose mean fixed the shift in Eq. (3). One worldline ensemble, run once, yields both numbers. Section IV C validates this identity against a live pipeline run.

Two constructions of the ensemble average must be told apart, because one of them silently destroys the visibility signal. Averaging the per-worldline phases first, and only then taking the phasor, $|\exp(i\langle\Delta\Phi\rangle)|$, gives modulus 1 for any spread in the underlying phases. That happens because the spread is discarded before the modulus is ever taken. The correct order, Eq. (4), sums the phasors themselves and takes the modulus of that sum, which is what lets a real phase spread show up as $V < 1$. Both orderings are encoded as opposing kill tests in the pipeline's own test suite (`tests/test_coherent_visibility.py`), so a future change that collapses the two back together is caught mechanically.

III. THE ADDITIVE TABLE AS THE FIRST-ORDER EXPANSION

The standard additive budget turns out to be the first term of the product rule's own expansion, and every higher-order term it drops is small enough to ignore at current clock precision. Expanding the product in Eq. (1) to first order in each p_k recovers the additive budget exactly:

$$\prod_k (1 + p_k) \;=\; 1 + \sum_k p_k \;+\; \sum_{j<k} p_j p_k \;+\; \dots \tag{5}$$

The standard table, one row per systematic, rows summed linearly, is the truncation of Eq. (5) at first order. Every cross term the truncation drops is a product of two or more systematics that are each already small, and at the field's current performance those products are far smaller still. The pipeline's own composition rules state each bound directly, as part of the codebase's own documentation. The DC-Stark and blackbody-radiation terms are independent scalar perturbations of the atom's internal energy. Their second-order cross term averages to exactly zero for an isotropic thermal field, an exact cancellation. Their multiplicative product correction, the Stark×BBR cross term of Eq. (5), is bounded at order 10^{-33}.[2] The gravitational-redshift term depends only on an atom's position in the local potential, and never on its motional or internal state, so its cross term with any field-driven shift is bounded at order 10^{-31}.[3] Both bounds sit at least twelve orders of magnitude below the 10^{-18}-to-10^{-19} systematic floor the clocks of Sec. I report.

This comparison settles what the product form is for. It states one composition rule that explains why the additive table works as well as it does. It also gives that rule's own quantitative boundary, the order at which it would stop working, in the same units the rows themselves are reported in. The additive budgets of Sec. I are already correct to a precision no experiment now approaches. This paper does not correct any of those published totals, and revises no published number. The three cases of Sec. IV show something different: where the product form's other structural consequences change a conclusion. A nonlinear function of a distributed quantity gets evaluated before it is averaged; two systematics compose as correlated factors on one shared rate; a fringe visibility gets read off the same object that sets the shift. None of the cross-term corrections above ever enters that arithmetic.

IV. THREE CONSEQUENCES OF COMPOSING ON ONE RATE

A. Per-state evaluation before averaging: a multi-surface thermal environment

A blackbody-radiation budget row is ordinarily a single number: a lab characterizes its enclosure by one effective temperature and evaluates a Planck-integral shift formula at that one temperature. A clock deployed outside a temperature-stabilized laboratory chamber, in a vehicle or behind a viewport a diurnal cycle warms, sees an enclosure whose visible surfaces can differ by tens of kelvin. The blackbody shift's dynamic term scales as T^6 through T^{10}, higher powers than the T^4 power the static shift and the naturally chosen effective temperature both track. A single temperature can be chosen to match the static (T^4) moment of a non-uniform environment. That choice generally misses the environment's higher dynamic moments, and the companion paper quantifies the resulting mismatch. It

[2] docs/CONVENTIONS.md, section 13, "(E33) Scalar pivot composition."
[3] Companion paper, Sec. II.E, "The gravitational-redshift pivot term and extended samples."

TABLE I. Reconciling the published Al^+ secular-motion total: the pipeline's full treatment, the bold final step, lands inside Marshall et al.'s published band at 0.08σ. The five steps above it are deliberately incomplete treatments of the same published six-mode inputs [5], read top to bottom, each adding one piece of physics the previous step omitted; their verdicts are kept to show what each omission costs, and step 1's agreement is the accidental near-cancellation the text explains. Copied from the companion paper's own progression table and Fig. 8 there. σ is the distance from Marshall et al.'s published band, combining the published and predicted uncertainties in quadrature. Participation and micromotion enter as two multiplicative factors on the same per-mode rate (Eq. (1)).

Step	Treatment	σ	Verdict
1	Single species mass, every mode	0.10	MET
2	Participation, mass-ratio closed form	14.11	NOT MET
3	Participation, measured-spectrum fit	14.01	NOT MET
4	Participation × leading-order micromotion	1.62	NOT MET
5	Participation × exact-Floquet micromotion	1.62	NOT MET
6	**Coupled two-ion Floquet, constrained fit**	**0.08**	**MET**

crosses 10^{-18}, comparable to a modern lattice clock's total systematic budget, at roughly an 11 K spread across two comparably sized surfaces, and at roughly 19 K for a small warm viewport against a larger chamber body.[4]

The pipeline's multi-surface extension evaluates the blackbody shift's moments per surface, before they are combined into a single number. Writing w_i for each surface's solid-angle fraction and T_i for its temperature, the environment's n-th weighted moment is $M_n = \sum_i w_i(T_i/T_0)^n$. The static and dynamic terms of the shift formula are each evaluated against their own moment M_n, in place of a single power $(T/T_0)^n$ of one effective temperature.[5] A uniform, single-surface environment reduces this construction to the ordinary single-T formula term for term: the two paths share the same coefficient-evaluation code and agree at floating-point precision. The construction is validated against a real transportable clock's own position-resolved measurement. Replaying Nosske et al.'s published aperture geometry, interior emissivity, and shield temperatures through the pipeline's enclosure-and-aperture model predicts a differential shift, as the atoms move from the shield's center out through an aperture, of -3.325×10^{-15}. Nosske et al.'s own closed-form model gives $-3.32(7) \times 10^{-15}$ for that same shift, and their own measurement gives $-3.33(3) \times 10^{-15}$. The pipeline's prediction lands within 5.011×10^{-18} of their model and 4.989×10^{-18} of their measurement, well inside either one's own stated uncertainty.[6]

The point generalizes past blackbody radiation. A single effective value often stands in for a physically distributed quantity throughout an optical-clock budget: an effective lattice depth over a trapped ensemble's spatial extent, an effective field over a cloud's own spread. Each substitution risks the same failure mode a single effective temperature shows here. A nonlinear function of the distributed quantity, evaluated at one collapsed value, does not equal that same nonlinear function's own moments evaluated across the real distribution and then combined. The composition rule of Sec. II states the same discipline once, for every systematic and every ensemble the pipeline evaluates. The distribution comes first. The moments, the mean that sets the shift and the variance that sets the visibility, come from integrating over the full distribution, before any collapse to one representative value.

B. Two correlated systematics as factors on one rate: reconciling a trapped-ion secular-motion budget

A trapped two-ion clock's motional time-dilation shift depends on two physically distinct pieces of the same underlying motion. One piece is how much of each shared vibrational mode the clock ion itself carries, its participation. The other is how much extra oscillation, at the trap's own radio-frequency drive, that secular motion itself induces, its intrinsic micromotion. A budget table built one row at a time can compute either piece on its own. Marshall et al.'s published $^{27}Al^+/^{25}Mg^+$ two-ion crystal is a real case where doing so goes badly wrong before it goes right. Table I reproduces the companion paper's own six-step progression through this reconciliation. Each row is built from Marshall et al.'s own published six-mode frequencies and occupations, and reported against Marshall et al.'s own published secular-motion band of $-114.6(3.8) \times 10^{-19}$.[7]

Step one carries one species mass through every one of the six modes, with no participation split and no micromotion correction. It lands inside Marshall et al.'s own published band at 0.10σ. That agreement is a near-cancellation of

[4] Companion paper, Sec. II.D, "The multi-surface thermal environment."
[5] docs/CONVENTIONS.md, section 13, "(E37) Multi-surface thermal environment."
[6] Companion paper, Sec. IV.J, "The PTB position-resolved thermal-environment validation," citing Nosske et al. [6].
[7] Companion paper, Secs. IV.H-I, "The Al-ion secular-motion arithmetic reproduction" and "Per-mode participation and the intrinsic-micromotion enhancement"; docs/CONVENTIONS.md section 16.

two large omissions moving in opposite directions. The true clock-ion participation in a shared mode is close to one-half. The true intrinsic-micromotion enhancement on the radial modes is close to a factor of two. The single-mass step's implicit choice of one for both factors happens to land close to their true product. That is an accident of the approximation, one the next two steps expose. Supplying only the participation correction, steps two and three, removes that accidental cancellation without replacing it. The resulting total under-predicts the published band by a factor of close to two, a 14σ discrepancy against Marshall et al.'s own stated uncertainty. Composing the intrinsic-micromotion enhancement back in as a second multiplicative factor on the same per-mode rate, steps four and five, closes most of that distance, to 1.62σ. That result still falls outside the published band. The remaining gap traces to a specific mechanism the companion paper identifies: the two-mode pairs sharing an axis run at Coulomb-shifted collective frequencies a single-ion enhancement factor cannot see. Solving the two ions' coupled equations of motion, then fitting the resulting two-parameter trap model to all four measured radial frequencies at once, closes the budget to 0.08σ, step six, comfortably inside Marshall et al.'s own published band.

A budget table computes participation and micromotion as separate rows, or omits the smaller of the two once a single-mass evaluation already looks close to the published value. Composing them as two correlated factors on the same per-mode rate exposes the near-cancellation that single-mass evaluation hides. It also lets the coupled two-ion physics that resolves it enter as one more correction to that shared rate, the same object every other correction in this section already multiplies.

C. Shift and fringe visibility from the same phase distribution

Section II states that the same worldline ensemble that sets the mean fractional shift, Eq. (3), also sets the Ramsey fringe visibility, Eq. (4). The visibility calculation uses the same accumulated-phase draw as the shift calculation, with no free parameter of its own. The companion paper validates the Gaussian closure $V = \exp(-\sigma_\Phi^2/2)$ this identity predicts against a live pipeline run. The run sweeps a classical worldline ensemble's squeezing parameter over several values, at 4000 worldlines each. At each value, the pipeline's own reported visibility is compared against the same run's independently computed Gaussian-closure check. The largest relative deviation between the two, at the largest squeezing swept, lands at roughly 0.65%. That deviation is consistent with the $O(\sigma^4/N)$ Monte Carlo scaling the pipeline's own test-tolerance derivation documents for this comparison.[8] That number comes from the same accumulated-phase distribution Eq. (3) already builds to report the mean shift, with the same worldline draw and the same weights supplying both outputs.

V. RELATION TO PRIOR PRACTICE

Relativistic timing already composes a rate scalar for a clock's own systematics, for the terms general relativity itself supplies. Ashby's treatment of relativity in the Global Positioning System expresses each GPS satellite clock's rate, relative to a ground clock, as a product of the gravitational potential term and the special-relativistic velocity term. The two compose multiplicatively along the satellite's worldline, the same proper-time-rate object this paper composes for an optical clock's atomic-environment terms [7]. Chronometric geodesy extends the same idea in the other direction. An optical clock's own measured rate, relative to a reference, is read as a measurement of the gravitational potential difference between the clock and the reference. A recent review of atomic clocks for geodesy surveys this practice across a range of clock platforms and baselines, each one treating the clock's fractional frequency as a rate factor tied to the local metric [8].

Both of these established practices compose rate scalars for the gravitational and kinematic terms general relativity itself supplies. This paper's own claim is about scope. What CliffordClock adds is the same composition, extended to a clock's full budget. That budget includes atomic-environment terms with no counterpart in a GPS satellite's or a geodetic baseline's own rate factor: the DC-Stark shift, blackbody radiation, and the electric-quadrupole shift. Those terms are evaluated per quantum state where the physics calls for it, the per-surface thermal moments of Sec. IV A and the per-mode motional factors of Sec. IV B among them. Fringe contrast comes out of that same object, Sec. IV C, as a byproduct of the shift calculation. The implementation is open, and every claim above is checked against a published budget in the repository's own test suite. No claim is made that this is the only way to compose a clock's systematics, or the first time a rate scalar has been composed for any of its individual terms. The claim that is made is about bookkeeping. One rate scalar per atom carries the whole budget: every effect enters it the same way, and

[8] notebooks/13_trapped_ion_quantum_motion.ipynb, Sec. 5, "Ramsey visibility under squeezing"; companion paper, Sec. II.H, "Coherent Ramsey visibility," Eq. (E39).

adding an effect means adding one factor. Effects that share the same underlying motion stay correlated, because they multiply the same per-mode rate. That correlation is what closed the budget of Sec. IV B. The shift and the fringe contrast come out of one calculation. The full rule fits in the listing of the next section, short enough to check by reading it.

VI. THE COMPOSITION, IN SIX LINES

Every equation of Sec. II, the multiplicative rate, the phase integral, the shift as a mean, and the visibility as a modulus of the mean phasor, is this short:

```
def clock_phase(worldline, terms, nu_0):
    rate = lambda t: nu_0 * prod(1.0 + p(worldline.state(t)) for p in terms)
    return 2.0 * pi * integrate(rate, worldline.proper_time)

phases = [clock_phase(w, terms, NU_0) for w in ensemble]
shift = mean(phases) / (2.0 * pi * NU_0 * T) - 1.0
visibility = abs(mean(exp(1j * array(phases))))
```

Each entry of `terms` is one systematic's own physics, computed from its own formalism, as any published budget row is; the composition is everything else in the listing. The source code, test suite, and the companion validation paper underlying every number in this paper are available at `https://github.com/velar-mbr/CliffordClock` under the GNU Affero General Public License v3.0 or later (AGPLv3).

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